Chapter 13 • Probability

Solved Review Exercise 13

MCQs

1. Each element of a sample space is called:
a) event
b) experiment
c) sample point
d) outcomes
Ans: c
An individual element is a sample point.

2. An outcome representing how many times we expect something to happen is called:
a) outcomes
b) favourable outcome
c) sample space
d) sample point
Ans: b
Printed key: (b). Precise terminology is “expected frequency”, which is missing; a favourable outcome is simply an outcome in the event.

3. How often an event occurs relative to total trials is:
a) expected frequency
b) sum of relative frequency
c) relative frequency
d) frequency
Ans: c
Relative frequency=event frequency/total trials.

4. Estimated probability from observed trials is also known as:
a) relative frequency
b) expected frequency
c) class boundaries
d) sum of expected frequency
Ans: a
Observed relative frequency estimates probability.

5. The sum of expected frequencies equals the fixed number of:
a) trials
b) relative frequencies
c) outcomes
d) events
Ans: a
$\Sigma N \cdot p = N$ when probabilities sum to 1.

6. The chance of occurrence of an event is called:
a) sample space
b) estimated probability
c) probability
d) expected frequency
Ans: c
This is the definition of probability.

7. An event with a greater chance to occur is called:
a) equally likely event
b) likely event
c) unlikely event
d) certain event
Ans: b
A likely event has high, but not necessarily certain, probability.

8. Total sample-space size when four dice are rolled is:
a) 6^2
b) 6^3
c) 6^4
d) 6^6
Ans: c
Each of four dice has six outcomes: $6^4=1296$.

9. For a pair of dice, P(double 2) is:

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a) $\frac{1}{6}$
b) $\frac{1}{3}$
c) $\frac{5}{6}$
d) $\frac{1}{36}$

Ans: d

Only (2,2) among 36 ordered outcomes.

10. For a 52-card pack, P(no jack and no king) is:

a) $\frac{2}{13}$
b) $\frac{11}{13}$
c) $\frac{2}{52}$
d) $\frac{11}{52}$

Ans: b

Exclude four jacks and four kings: $\frac{44}{52}=\frac{11}{13}$

2. Define relative frequency and expected frequency.

(i) Relative frequency=$\frac{\text{frequency of a particular event}}{\text{total frequency or trials}}$; it estimates probability from observations.
(ii) Expected frequency=N×P(A), the predicted count of event A in N trials under the stated probability model.

3. An urn contains 10 red,5 green and8 blue balls (23 total).

(i) Green=$\frac{5}{23}$
(ii) red=$\frac{10}{23}$
(iii) blue=$\frac{8}{23}$
(iv) Not red=$\frac{5+8}{23}=\frac{13}{23}$
(v) not green=$\frac{10+8}{23}=\frac{18}{23}$

4. Three coins are tossed together.

There are $2^3=8$ equally likely outcomes. Head counts 0,1,2,3 occur in 1,3,3,1 outcomes respectively.

(i) Exactly three heads: $\frac{1}{8}$
(ii) At least two tails means 0 or1 heads: $\frac{1+3}{8}=\frac{1}{2}$
(iii) Not at least two heads also means 0 or1 heads: $\frac{1}{2}$
(iv) Not exactly two heads: $1-\frac{3}{8}=\frac{5}{8}$

5. A card is drawn from a 52-card pack.

(i) Red king or red jack: two red suits×two ranks = 4 cards, so $\frac{4}{52}$
$=\frac{1}{13}$
(ii) Not the 2 of clubs and not the 2 of spades: exclude those two cards, so $\frac{50}{52}=\frac{25}{26}$

6. Six coins are tossed 600 times; tail-count frequencies are 110,90,105,80,76,123,16 for 0–6 tails.

Total=600. Relative frequency=each $\frac{\text{frequency}}{600}$

Relative frequencies are tabulated below; their sum is 1.

Number of tails	Frequency	Relative frequency	Decimal
0	110	11/60	0.18333
1	90	3/20	0.15000

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Number of tails	Frequency	Relative frequency	Decimal
2	105	7/40	0.17500
3	80	2/15	0.13333
4	76	19/150	0.12667
5	123	41/200	0.20500
6	16	2/75	0.02667
Total	600	1	1.00000

7. A lot of 25 items contains 8 defective items. Find relative and expected frequency of non-defective items.

Non-defective items = 25−8
= 17. Relative frequency
= $\frac{17}{25}$
= 0.68.

For this single lot, expected non-defective count=$25\times\frac{17}{25}$=17 items.

Relative frequency=0.68; expected frequency=17 non-defective items.

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